

Dynamic Context Distillation: Resilient Foundation Models for Decentralized Traffic Forecasting

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Background

Accurately forecasting highway traffic speeds is essential for efficient daily urban mobility and real-time crisis management. To remain resilient during sudden disruptions or data shifts, predictive systems must be able to adapt, rendering slow, sensor-specific training pipelines impractical. New foundation models like Chronos-2 offer state-of-the-art zero-shot accuracy alongside intrinsic uncertainty estimation [1]. However, the quadratic scaling bottleneck inherent to transformer architectures forces redundant historical processing at every inference step, demanding stable cloud connectivity and limiting deployment on resource-constrained edge devices.

Methodology

To overcome this bottleneck, we propose a distillation pipeline that compresses complex historical traffic patterns into small efficient LoRA adapters. Inspired by the work of Charakorn et al. [2], we train a meta-learning hypernetwork to dynamically generate these adapters in a single forward pass. Optimization is driven by minimizing the divergence between the student model and either a full-context teacher distribution or observed ground-truth targets. We empirically validate this architecture on the PEMS-BAY dataset, predicting highway speeds across a complex network of 325 sensor stations in San Francisco.

Keywords: Meta-learning, Transfer Learning, LoRA, Transformers, Uncertainty Estimation

Recommendations

We recommend deploying this lightweight architecture on resource-constrained edge nodes to provide highly resilient, low-latency traffic forecasting zero-shot.

Project Highlights

- We successfully compressed meaningful traffic patterns into lightweight LoRA adapters of just 590K parameters, extracting temporal intelligence for the 119M-parameter Chronos-2 foundation model while bypassing the memory and latency bottlenecks of full-context processing.
- We demonstrated that our adapted model achieves state-of-the-art forecasting performance (measured in MAE) under constrained context lengths visualized in Fig. 1.
- Our approach exhibited robust transferability: training the hypernetwork on a sparse subset of just 20% of the PEMS-BAY sensor stations maintained comparable predictive accuracy across the remaining 80% of the stations.
- We validated the adaptability of our meta-learning framework by generating effective LoRA adapters for an entirely unseen transportation network (METR-LA dataset),

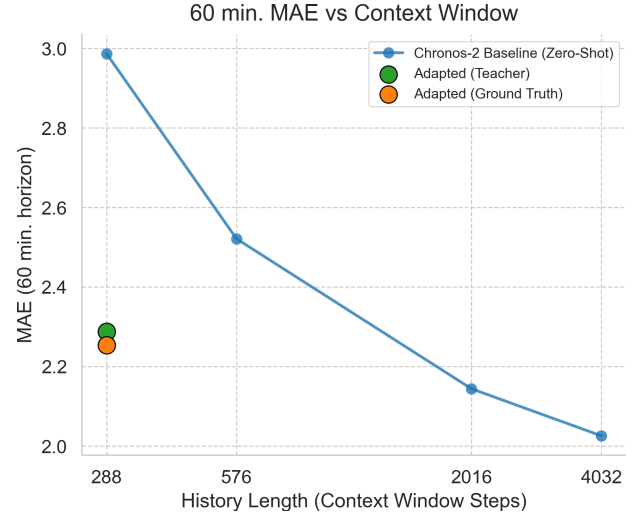


Fig. 1. **60-minute forecasting error (MAE) across context lengths.** Using just 288 context steps, our adapted models (green and orange) significantly outperform the zero-shot baseline (blue) at both 288 and 576 steps, nearly matching its accuracy at 2016 steps. 288 steps corresponds to 24 hours.

yielding consistent forecasting performance and proving the method’s reliability in novel, unmapped urban areas.

- We verified that the foundation model’s uncertainty estimation remains intact under the context compression, evaluated via empirical coverage.

Limitations

- The hypernetwork may over-rely on the strict 24-hour periodicity of the PEMS-BAY dataset. This risks poor generalization to non-seasonal, volatile domains like retail demand or financial anomalies.
- The framework cannot process external covariates like weather or holidays. This restricts its real-world commercial utility, as it remains blind to exogenous data.
- While the resulting LoRA adapter is lightweight, the training phase of the hypernetwork requires simultaneously loading the teacher model, context encoder, and hypernetwork leading to a high memory footprint.

[Full code available.](#)

REFERENCES

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